Math 128a: Midterm #2

Nov 19 - Nov 21

YOUR NAME:	STUDENT ID:

1	2	3	total
8 pts	8 pts	8 pts	24 pts

Please read the following instructions carefully:

- This exam is 80 minutes (+5 minutes to scan and upload solutions). If you have DSP accomodations for additional time, make sure your time has been extended on Gradescope.
- The exam is available starting on Wednesday, November 19, at noon PST. **The deadline to complete it is Friday, November 21, at 11:59 PM PST.** You may take it at any time from when it is available till when the deadline is reached. Once you open the assignment on Gradescope, time will begin. Your time will end when your exam time is up or the deadline is reached, whichever is sooner. If you start the exam too close to the deadline, you might not have the full exam duration to take it.
- You may use course materials (anything available on bCourses, personal notes, and the textbook). It is **NOT** open internet, and communication about the exam is **NOT** allowed until Saturday, Nov 22. Calculators and MATLAB are **NOT** allowed or required.
- If you have questions about the exam, you may post a **PRIVATE** message on Ed. Staff will be online during Thursday's classtime. You can expect reasonably quick answers then. If you take it at any other time, response times could be longer.
- You may only solve the exam on physical paper or on a note-taking app (i.e., handwritten), unless you have specific DSP accommodations stating otherwise.
- Clearly indicate the number and part of the problem you are solving on the corresponding answer sheet.
- When you take the exam, make sure that you are somewhere with internet, paper and pens, and **a device to take pictures** to upload your solution.

Good luck!

Problem 0: Academic Integrity

Please copy the following statement in your own handwriting, and then sign your name below it.

> *I affirm that I have read and will follow all instructions. I will not cheat, and I will not violate academic integrity. As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Problem 1: Nonlinear equations [8 points]

(a) Find an algebraic expression for the unique root (in $\mathbb{R}$) of $f(x) = x^2 - 2/x$.

(b) Newton's method attempts to find a root using the iteration $p_{n+1} = g(p_n)$. Find $g(x)$. Simplify your answer.

(c) Is the convergence linear? Is it at least quadratic? Justify your answer.

(d) Bonus (1 point): Find the exact order of convergence.

Answer of Problem 1

(a) The root of f is $2^{1/3}$.

(b) Newton's method is $p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$, so $g(x) = x - \frac{f(x)}{f'(x)}$. Since $f'(x) = 2x + 2x^{-2}$, we obtain

$$g(x) = x - \frac{x^4 - 2x}{2(x^3+1)} = \frac{2x(x^3+1) - x^4 + 2x}{2(x^3+1)} = \frac{x^4 + 4x}{2(x^3+1)}.$$

(c) We have

$$g'(x) = \frac{(x^3-2)^2}{2(x^3+1)^2},$$

thus $g'(2^{1/3}) = 0$ and Theorem 2.8/2.9 implies that the iteration converges at least quadratically near the fixed point.

(d) We need to compute more derivatives of g to figure this out. We have

$$g''(x) = \frac{9x^2(x^3-2)}{(x^3+1)^3},$$

and therefore $g''(2^{1/3}) = 0$, hence the convergence is at least cubic. For the third derivative, we have

$$g^{(3)}(x) = \frac{9x(-4x^6 + 19x^4 - 4)}{(x^3+1)^4},$$

which is clearly non-zero at the root, therefore the convergence is cubic but not quartic.

Problem 2: Polynomial interpolation [8 points]

Eric and Giuseppe collect some data. They get $f(0) = 0$, $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, $f(4) = 16$, and $f(5) = 25$. Eric uses Neville's method to compute a minimal degree polynomial $N(x)$ that passes through those six points. Giuseppe uses the Lagrange basis to compute a minimal degree polynomial $L(x)$ that passes through those six points. They both fully simplify their expressions.

(a) What is Eric's final expression for $N(x)$? You need NOT reproduce all his calculations.

(b) What is Giuseppe's final expression for $L(x)$? You need NOT reproduce all his calculations.

(c) Compute the divided difference $f[0, 1, 2, 3, 4, 5]$. Hint: Tables are unnecessary.

(d) Recall that $\pi \approx 3.14 + 0.00159265$. Isaac collects an additional datapoint $f(\pi) = 3.14^2$. He writes down an interpolating polynomial $P(x)$ for the combined seven points. Using the recursive form of Newton's divided difference formula, write down an expression for $P(x) - L(x)$ in terms of divided differences and a factored polynomial.

(e) Use your answers in parts (b) and (d) to compute the divided difference

$$f[0, 1, 2, 3, \pi, 4, 5].$$

Your answer should be an algebraic expression in terms of π.

(f) It is well-known that polynomial interpolation does poorly in regions that are far away from the input data. What is the limit of $P(x)$ as $x \to +\infty$? In one sentence, explain your reasoning.

Answer of Problem 2

(a) From inspection, it is clear that x^2 passes through all the points. By uniqueness of the polynomial interpolation, the result of Neville's method is $N(x) = x^2$.

(b) For the same reason, the result of Lagrange interpolation is $L(x) = x^2$.

(c) The divided difference $f[0,1,2,3,4,5] = 0$ is the coefficient of x^5 in $L(x)$.

(d) Newton's divided difference formula gives us

$$P(x) - L(x) = f[0,1,2,3,\pi,4,5]x(x-1)(x-2)(x-3)(x-4)(x-5).$$

In other words, the difference between the polynomials $P(x)$ and $L(x)$ is a multiple of $x(x-1)(x-2)(x-3)(x-4)(x-5)$. Recall that $x(x-1)(x-2)(x-3)(x-4)(x-5)$ is a polynomial that is zero at 0, 1, 2, 3, 4, and 5. Multiples of this polynomial will not affect the value at those points.

(e) It remains to determine which multiple of $x(x-1)(x-2)(x-3)(x-4)(x-5)$ produces the perturbation necessary to make $P(\pi)$ pass through the remaining point $f(\pi) = 3.14^2$. We can do this by plugging π into our previous formula.

$$P(\pi) - L(\pi) = f[0,1,2,3,\pi,4,5]\pi(\pi-1)(\pi-2)(\pi-3)(\pi-4)(\pi-5).$$

Note that $P(\pi) = 3.14^2$ because it interpolates the additional point and that $L(\pi) = \pi^2$ because $L(x) = x^2$. We get

$$f[0,1,2,3,\pi,4,5] = \frac{3.14^2 - \pi^2}{\pi(\pi-1)(\pi-2)(\pi-3)(\pi-4)(\pi-5)}.$$

(f) Observe that $L(x) = x^2 \to +\infty$ in the limit as $x \to +\infty$. When we add an additional node, we add higher order terms to our polynomial. These higher order terms can be thought of as small corrections in regions where you are close to many nodes, but they dominate when you are far away from sufficiently many nodes. This is why polynomial interpolation can look strange near the boundary of an interval with uniform nodes and why you might prefer Chebyshev points, which ensures that you are always close to sufficiently many nodes.

In this problem, the higher order terms dominate because we are looking at the regime $x \to +\infty$. This means that $P(x) = f[0,1,2,3,\pi,4,5]x^6 + O(x^5)$. Higher powers are larger here, so $O(x^5)$ is smaller than x^6. It suffices to check the sign of the divided difference. By inspection, the numerator is negative, and the denominator is positive. As a result, $P(x) \to -\infty$. The answer to this problem was determined by whether the additional point in $P(x)$ was above or below the existing polynomial $L(x)$ as well as the parity of the number of nodes to the right of the added point.

Answer of Problem 2

Problem 3: Numerical integration [8 points]

Consider the function

$$f(x) = \begin{cases} h(x)^3 + A & -2 \le x < -1/4 \\ A & -1/4 \le x < 2 \\ g(x) & 2 \le x \le 4 \end{cases}$$

where $h(x)$ and $g(x)$ are smooth, and $h(-1/4) = 0$.

(a) Compute the value of the integral using the composite midpoint rule and $n = 6$. Use the convention from the end of Lecture 20.

Solution.

$$\int_{-2}^{4} f(x)\,\mathrm{d}x = 2(f(-1) + f(1) + f(3)) = 2(h(-1)^3 + A) + 2A + 2g(3)$$

(b) Give and *justify* an error estimate for this integration. Be careful about which results from class you can apply and where you can apply them.

Solution. From lecture/the book, we know that composite midpoint error is given by

$$\frac{b-a}{6} h^2 f''(\mu)$$

as long as f is C^2 on (a, b). Therefore, we need to check on what intervals f is C^2.

- f is C^2 on the intervals $(-2, -1/4), (-1/4, 2), (2, 4)$.
- Clearly f is not necessarily C^2 at $x = 2$, since it may not even be continuous if $g(2) \neq A$.
- The other point we have to worry about is $x = -1/4$. We want to check if the left and right hand "limits" are equal. First

  $$h(-1/4)^3 + A = A$$

  so f is continuous at this point. Second

  $$\frac{d}{dx} h(x)^3 + A = 3h(x)^2 h'(x) \overset{x=-1/4}{\rightsquigarrow} 0 = \frac{d}{dx} A$$

  so f' is continuous. Third

  $$\frac{d^2}{dx^2} h(x)^3 + A = 6h(x)h'(x)^2 + 3h(x)^2 h''(x) \overset{x=-1/4}{\rightsquigarrow} 0 = \frac{d^2}{dx^2} A$$

  so f'' is continuous.

We conclude that f is C^2 on $(-2, 2)$ and $(2, 4)$. This means we can write an error estimate as the sum of two error estimates.

$$\frac{4}{6}(1^2) \max_{x \in [-2,2]} \left|6h(x)h'(x)^2 + 3h(x)h''(x)\right| + \frac{2}{6}(1^2) \max_{x \in [2,4]} |g''(x)|$$